

Urban Night Mobility Data Warehouse

NYC Citi Bike night mobility analysis with weather, holidays, and geographic enrichment

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- ❏ Full reproducible data warehousing pipeline built with PostgreSQL/PostGIS and Python.



Project Goal & Research Questions

Build a **reproducible data warehouse** to analyze night-time Citi Bike mobility in New York City during 2024, integrating weather, holidays, and geographic boundaries.

January 2024 is the validated development sample; the pipeline is designed for full-year execution.

Spatial Concentration

Where is night mobility concentrated across NYC boroughs?

Temporal Variation

How does night demand vary by weekday, weekend, and holiday windows?

Weather Impact

How do weather conditions affect night trip volumes?

Borough Flows

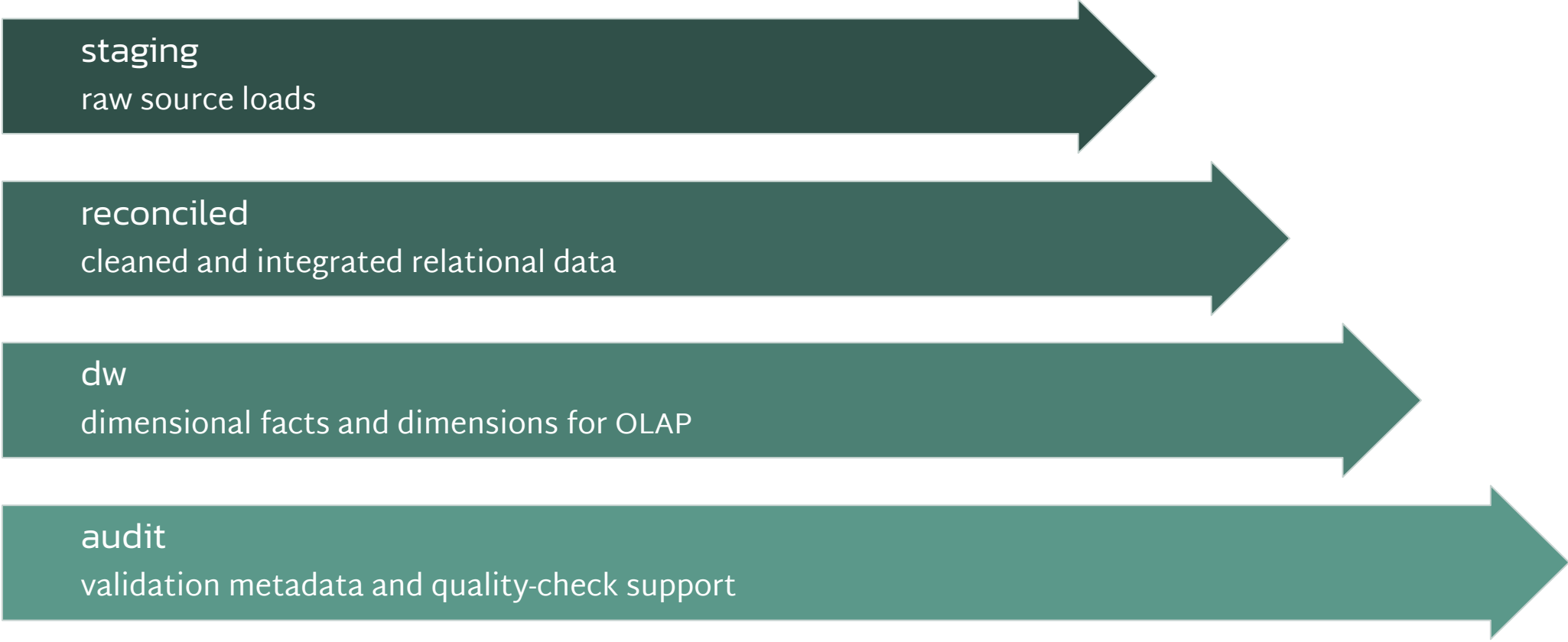
Which origin–destination borough flows are most relevant at night?

Four Heterogeneous Data Sources

Source	Content	Purpose
Citi Bike trip data	Trip-level bike-sharing records (start/end stations, timestamps, rider type)	Primary mobility fact source
NOAA daily weather (Central Park)	Temperature, precipitation, snow, and weather attributes	Enrich trips by environmental conditions
US public holiday calendar	US public holidays for 2024	Identify holiday and near-holiday mobility windows
NYC borough / NTA boundaries	Spatial polygons for boroughs and neighborhoods	Assign trips and stations to geographic zones via PostGIS

 Raw files are stored in `data_raw/` and ignored by Git to ensure reproducibility and manage storage.

End-to-End Pipeline Architecture



staging
raw source loads

reconciled
cleaned and integrated relational data

dw
dimensional facts and dimensions for OLAP

audit
validation metadata and quality-check support

ETL

ETL Operations & Reconciled Layer

Raw Sources

- CSV, JSON, and geospatial files
- Inconsistent timestamps and identifiers
- Incomplete or invalid trip records
- No geographic or holiday enrichment

Reconciled Layer

- Standardized timestamps and station identifiers
- Invalid or incomplete records cleaned or rejected
- Stations and trips integrated with borough geography
- Date attributes enriched: weekday, weekend, holiday flags
- Consistent relational tables ready for warehouse loading

ETL Implementation & Loading Flow



Raw Data Ingestion

Loading of Citi Bike CSVs, NOAA weather, holiday JSONs, and NYC geospatial files into staging.



Reconciled Integration

Timestamp standardisation, invalid trip filtering, station normalisation, geography assignment, and calendar enrichment.



Dimensional Warehouse

Surrogate-key lookup and fact table loading populate the star schema.



OLAP Analysis

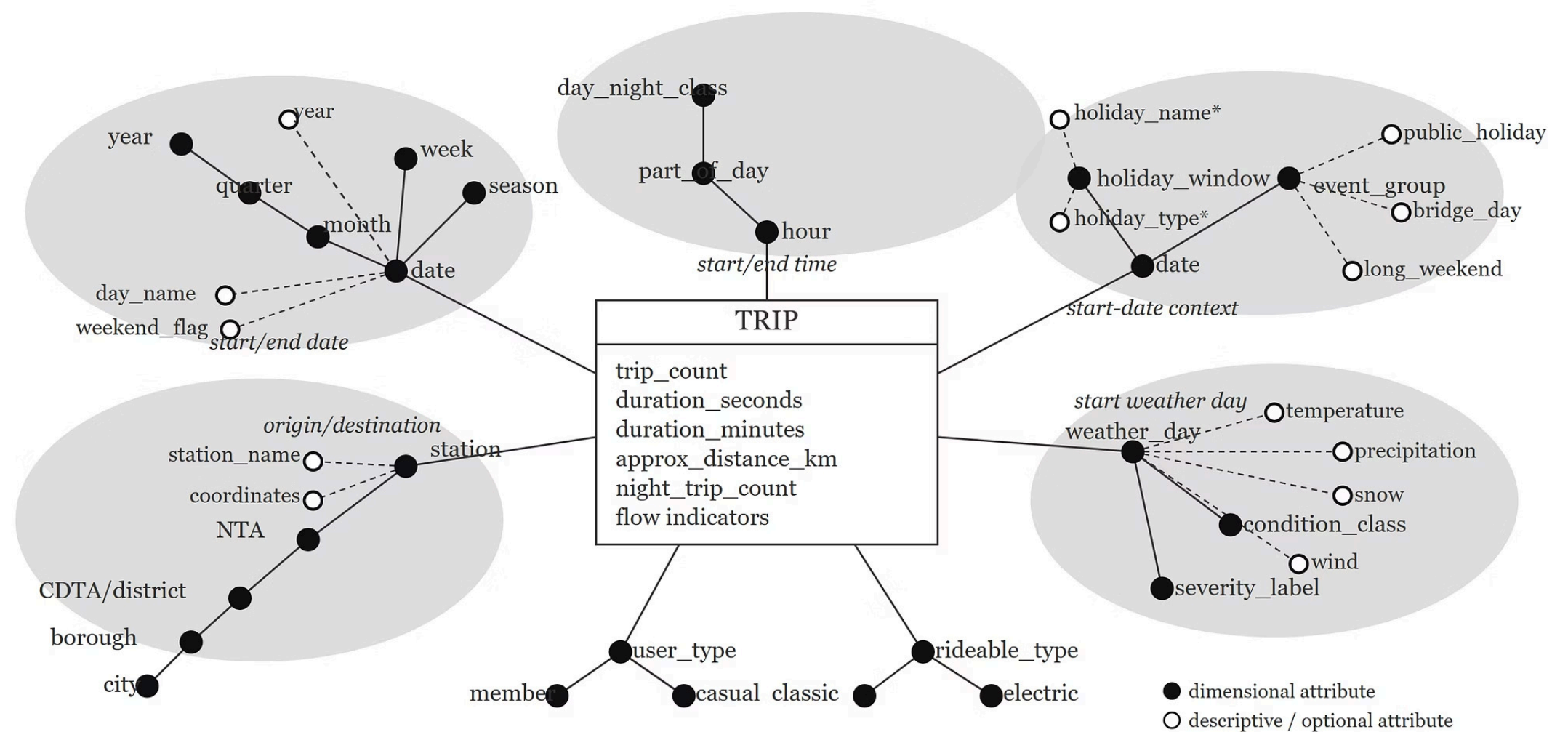
Data is ready for multi-dimensional querying and insights generation.

This sequential flow, primarily orchestrated with `Python` scripts and `PostgreSQL/PostGIS` operations, transforms heterogeneous raw sources into an analytically optimised dimensional model.

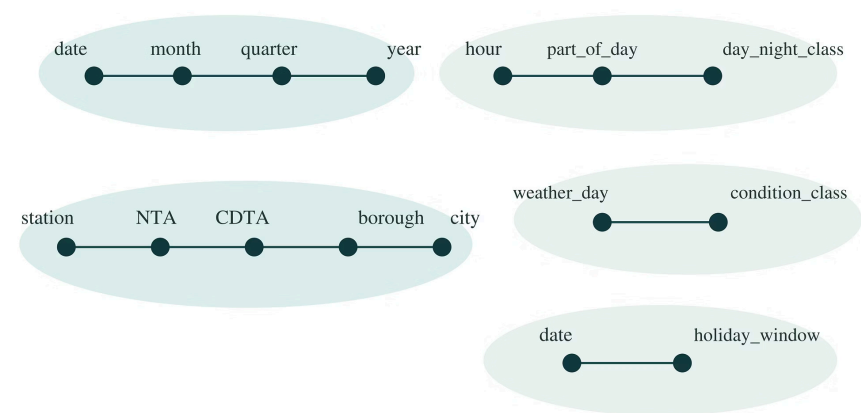
 Each stage incorporates validation and logging to ensure data integrity and traceability throughout the pipeline.

DFM Conceptual Model

At the conceptual level, the main fact is TRIP, with grain one accepted Citi Bike ride. The DFM is independent from PostgreSQL table names: it describes the fact, its measures, the analysis dimensions, their hierarchies, and the date-scoped contextual attributes used for OLAP.

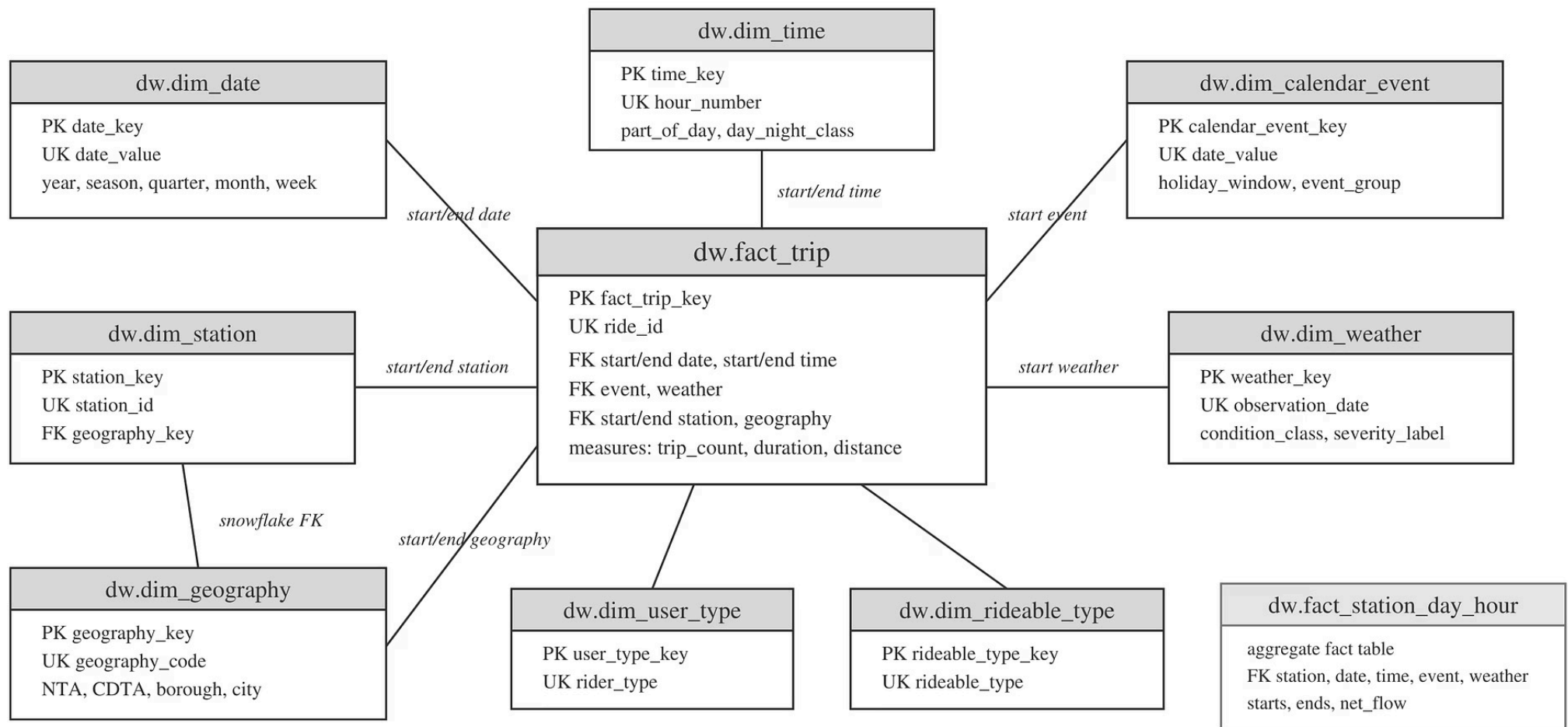


The role-playing dimensions are represented conceptually: Date, Time, Station, and Geography are used in start/end or origin/destination roles. Weather and calendar-event information are date-scoped contextual branches linked to the trip start context. Key hierarchies include:



Physical Snowflake Schema in PostgreSQL

The conceptual DFM is translated into the implemented ROLAP schema in PostgreSQL. The ride-grain fact table is `dw.fact_trip`. Dimension tables use surrogate primary keys and natural unique keys, while foreign keys connect facts to dimensions for OLAP querying.



The design is mostly star-shaped around `dw.fact_trip`, with a controlled snowflake edge from `dw.dim_station` to `dw.dim_geography`. Direct start/end geography keys are also stored in `dw.fact_trip` to support faster origin-destination roll-ups; this denormalization is validated through quality checks.

`dw.fact_station_day_hour` is an implemented aggregate fact table at station-date-hour grain (faster repeated station-flow OLAP)

Fact Tables

Our data warehouse incorporates two distinct fact tables, each designed to serve specific analytical needs by capturing different levels of granularity.

FACT_TRIP

Analytical Grain: One accepted Citi Bike ride.

Key Measures:

- `trip_count` (always 1)
- `duration_seconds` / `minutes`
- `approximate_distance_km`
- `night_trip_count`
- `member/casual` indicators
- `flow` contribution indicators

This granular fact table supports detailed, ad-hoc trip-level analysis, allowing deep dives into individual ride characteristics and direct linkage to specific events.

FACT_STATION_DAY_HOUR

Analytical Grain: Station + Date + Hour (hourly aggregations).

Key Measures:

- `starts` (number of trips starting from station)
- `ends` (number of trips ending at station)
- `net_flow` (ends - starts)
- `night_starts`
- `member/casual` starts

This aggregated fact table is optimised for faster station-level hourly analysis, significantly accelerating performance for common OLAP aggregations and trend identification.

By combining these two tables, we maintain the flexibility for detailed exploration (`FACT_TRIP`) while ensuring efficient performance for recurring aggregated queries (`FACT_STATION_DAY_HOUR`).

OLAP Query Design

Our OLAP queries extract analytical insights from the dimensional model, focusing on key aspects of urban night mobility. The SQL analyses use standard OLAP operations such as slice, dice, roll-up, drill-down, aggregation, and ranking.



Night Demand by Borough & Day Type

Analyse night-time Citi Bike demand across boroughs, slicing by night trips, rolling up to borough level, and dicing by comparing weekday, weekend, and holiday windows.



Weather Impact on Night Trips

Evaluate how weather conditions such as rain, clear weather, snow, and fog affect night-time trip volumes across weather classifications.



Borough Origin-Destination Flows

Identify the most frequent night-time borough flows by grouping trips into origin-destination pairs and ranking them by volume.



Station Hourly Net Flow

Use `FACT_STATION_DAY_HOUR` to examine hourly station inflow and outflow, revealing local imbalance and rebalancing pressure.



User Type Behaviour

Compare member and casual rider patterns by aggregating night demand across boroughs, day types, and time windows.



Rideable Type Usage

Compare classic and electric bike usage during night-time trips using trip volumes, durations, and spatial distribution.



Temporal Night Demand Patterns

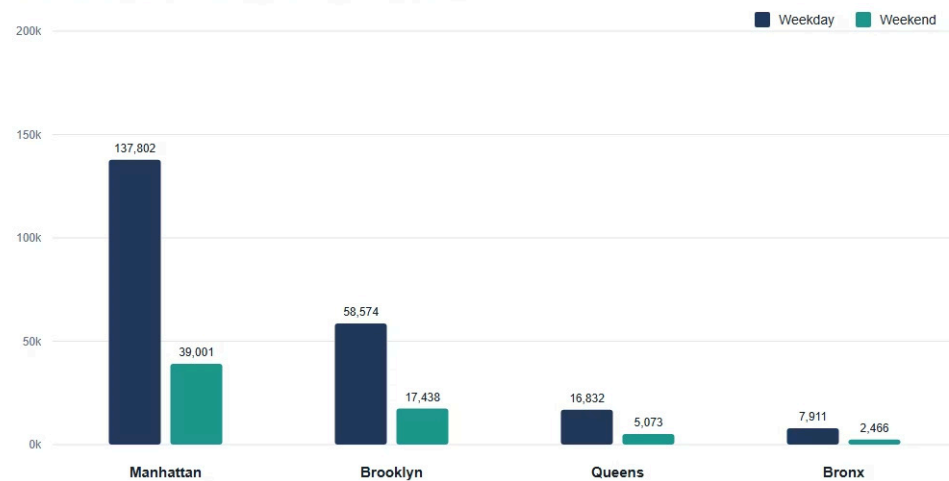
Drill down from calendar-level aggregations to hourly patterns to identify night mobility peaks across weekdays, weekends, and holiday windows.

All analytical queries are reproducible and stored in `sql/olap/01_olap_analysis.sql`.

OLAP Analysis: Night Demand by Borough & Day Type

Night trips by borough and day type

Slice on night trips; roll-up to borough; compare weekdays vs weekends.



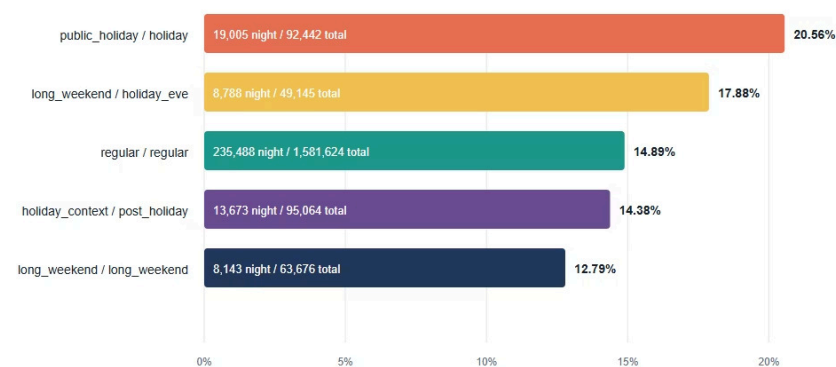
Source: dw fact and dimensions, January 2024 Citi Bike development sample

Key Finding

Night mobility is strongly concentrated in Manhattan, followed by Brooklyn.

Night-trip share by calendar context

Calendar-event dimension shows how holiday windows change relative night demand.



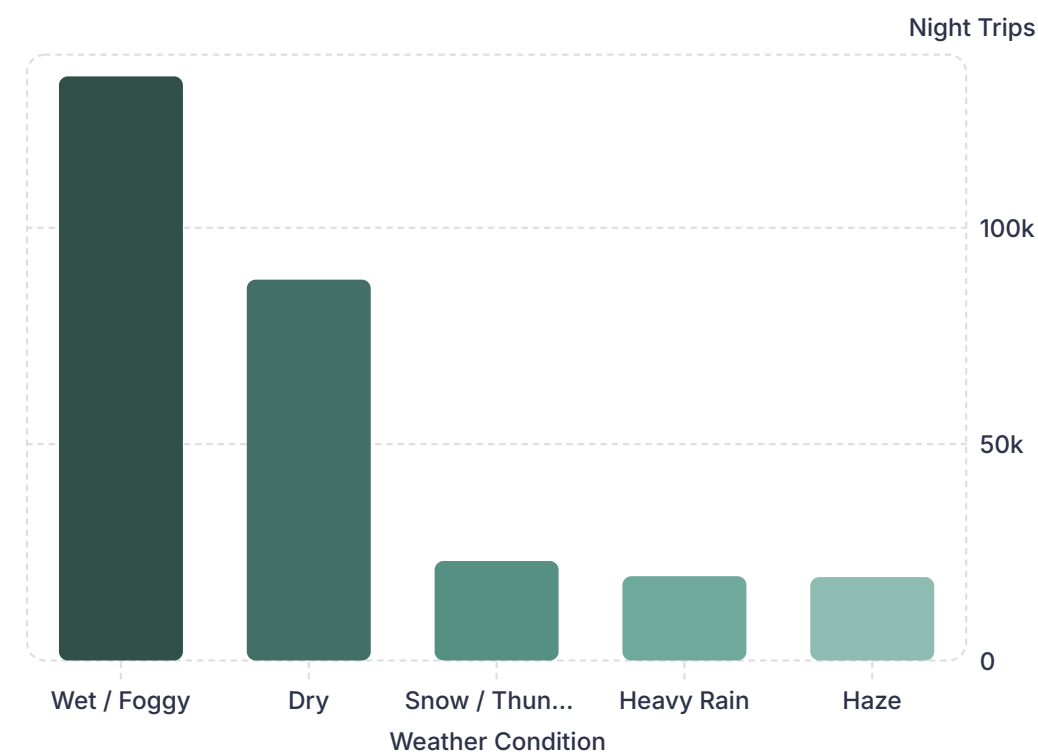
Source: dw fact and dimensions, January 2024 Citi Bike development sample

Key Finding

Holiday-driven night mobility rises mainly around public holidays and holiday eves, not long weekends.

Weather Impact & Borough Origin–Destination Flows

Weather & Night Trip Demand



Slice + aggregation + comparison: adverse weather significantly reduces night trip volumes. Weather enrichment connects mobility behavior with environmental context using the same fact table.

Top Borough OD Flows (January 2024)

Start Borough	End Borough	Night Trips
Manhattan	Manhattan	~95,000
Brooklyn	Brooklyn	~38,000
Manhattan	Brooklyn	~22,000
Brooklyn	Manhattan	~20,000
Queens	Manhattan	~12,000

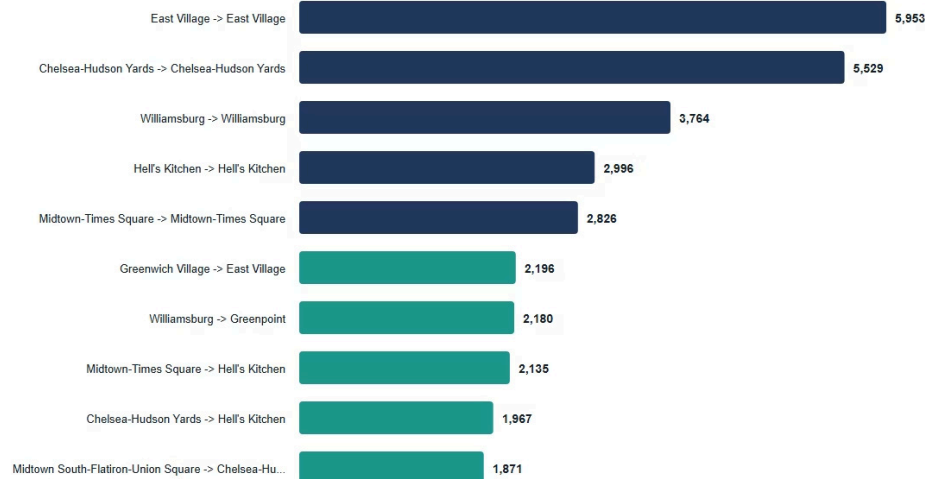
Roll-up + ranking: most night trips remain intra-borough. Inter-borough flows reveal key mobility corridors between Manhattan and Brooklyn.

Night Corridors and Station Imbalance

Top night origin-destination NTA corridors show that night mobility is mostly local and concentrated in dense mixed-use areas.

Top night origin-destination corridors

Slice on night trips; group by origin and destination NTA.

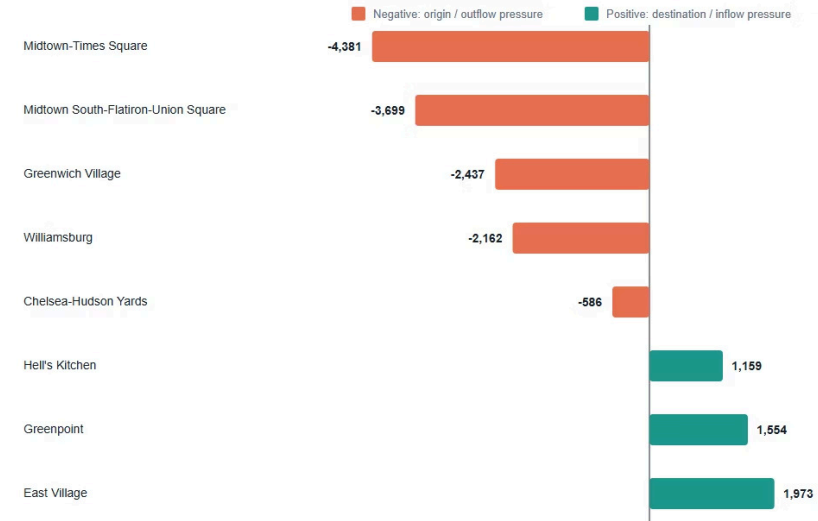


Source: dw fact and dimensions, January 2024 Citi Bike development sample

Station net-flow imbalance uses the aggregate fact table to reveal locations with stronger rebalancing pressure.

Night NTA net-flow for top corridor areas

Same NTA set as top night corridors; positive values mean more night arrivals than departures.



Source: dw fact_trip night trips, January 2024 Citi Bike development sample. Net-flow = night ends - night starts.

Night mobility is not only concentrated by borough; it also appears in local NTA corridors and station-level operational imbalances.

Main Findings from January 2024 Sample

Our initial OLAP analysis of the January 2024 sample has yielded several key insights into urban night mobility:

Night Demand by Borough & Day Type

Manhattan and Brooklyn are the dominant night-mobility boroughs, with weekdays consistently showing higher night demand than weekends.

Weather Impact

Adverse weather conditions significantly reduce night trip volumes, highlighting the sensitivity of demand to environmental factors.

Borough Origin–Destination Flows

Most night trips are intra-borough. The Manhattan–Brooklyn corridor represents the strongest inter-borough travel pattern.

Station Hourly Net Flow

Hourly station-level aggregations reveal dynamic local imbalance patterns between bike starts and returns, crucial for operational management.

User Type Behaviour

Distinct night-usage patterns observed between member and casual riders support the value of segmentation by rider type for targeted insights.

Rideable Type Usage

Analysis indicates varying night trip volumes, durations, and spatial distributions between classic and electric bikes, affecting overall mobility.

Temporal Night Demand Patterns

Recurring night-demand peaks have been identified across hours, weekdays, weekends, and holiday windows, informing predictive models.

Data Quality Controls

Quality checks are used to prove that the warehouse is reliable before interpreting OLAP results.

1. Source-to-Staging Validation

Raw source counts are compared with staging table counts for Citi Bike trips, weather rows, holidays, NTA boundaries, and borough boundaries.

2. Staging-to-Reconciled Accounting

Every Citi Bike row is accounted for: accepted trips + rejected trips = staging trips. Rejected records are not silently discarded; they are stored with explicit rejection reasons such as missing station identifiers or extreme duration.

3. Geographic and Temporal Coverage

Station-to-geography assignment is validated through PostGIS enrichment. Calendar and weather joins are checked for expected coverage.

4. Warehouse Integrity

Fact-to-dimension keys are checked across date, time, station, geography, weather, calendar, user type, and rideable type.

5. Aggregate Consistency

The station-day-hour aggregate fact is reconciled with the ride-grain fact to ensure totals remain consistent.

PROJECT OUTLOOK

Future Work

- Run the full-year 2024 pipeline to support seasonal and holiday-period comparisons.
- Add materialized views or further OLAP indexes for faster repeated analysis.
- Extend the presentation layer with dashboard-style outputs for the main OLAP queries.